MIL–D–13512B(MU)
24 MARCH 1964
SUPERSEDING
MIL–D–13512A(ORD)
28 APRIL 1961
(SEE 6.6)

MILITARY SPECIFICATION

DELAY ELEMENT, NON-DELAY M1 LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of delay element designated as Delay Element, Non-Delay, M1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–P–116 —Preservation, Methods of.
MIL–A–2550 —Ammunition and Special Weapons, General Specification for.
MIL–P–13392 —Primer, Percussion, M54.
MIL–I–45208 —Inspection Requirements, General Specification for.
MIL–I–45607 —Inspection Equipment, Supply and Maintenance of.

STANDARDS

MILITARY

MIL–STD–105D —Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–109 —Quality Assurance Terms and Definitions.
MIL–STD–1235 —Single and Multi-level Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

7548277 —Box, Packing, Ammunition, For Delay Elements, M1 and M2.
7548278 —Box, Set-Up, Packing, Amm., Inner and Outer For Delay Elements, M1 and M2.
8853675 —Delay Element, M1, Non-Delay Assembly.

PUBLICATIONS

ORDNANCE CORPS

IEL–8853675 —Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Material.** Materials and parts shall be

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in accordance with applicable drawings and specifications.

3.2 Delay elements. The delay element assemblies shall comply with all requirements specified on Drawing (dwg.) 8853675 and with all requirements specified in applicable specifications.

3.3 Functioning. The delay element assemblies shall not fail to function, and shall function with a maximum (max.) delay time of 0.003 seconds, when tested as specified in 4.4.2 (see 6.7).

3.4 Workmanship. All parts shall be free of chips, dirt, grease, rust, and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Contractor quality assurance system.* The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL–I–45208.

4.1.2 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.
(b) Number of units of product inspected.
(c) Results obtained for all inspection performed.
(d) Drawing, specification number and date, together with an identification and date of changes.
(e) Certificates of analysis on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.
(f) Number of items in the lot.
(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL–I–45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

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4.2 Initial production inspection.

4.2.1 *Submission.* Prior to the start of regular production, the contractor shall submit an initial production sample consisting of 150 assemblies and 25 sub-assemblies which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures, and equipment which will be used in fulfilling the contract. All parts and materials, including packaging, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analyses for materials. An initial production sample, or portion thereof, as directed by the contracting officer shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used or supply, as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer.

4.2.2 *Inspections to be performed.* Each assembly and subassembly shall be subjected by the Government to any or all of the examinations or tests specified in this specification and any or all requirements of the applicable drawings.

4.2.3 *Rejection.* If any assembly or subassembly fails to comply with any or all of the applicable requirements, the initial production sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any assembly or subassembly in the sample to comply with any or all of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new initial production sample or portion thereof. Until an initial production sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3 Inspection provisions.

4.3.1 *Lot formation.* A lot shall consist of delay element assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit as determined by the Government shall not necessitate changing the lot number. Product shall be submitted for inspection in accordance with Standard MIL–STD–105D (or Standard MIL–STD–1235, when applicable). Each lot shall contain:

(a) Parts and parts assemblies of one lot interfix number from one manufacturer.

(b) Primers of one lot interfix number from one manufacturer.

(c) Relays of one lot interfix number from one manufacturer.

4.3.2 *Examination.* Sampling plans and procedures for the following classification of defects shall be in accordance with Standard MIL–STD–105D except that inspection for Critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Primer holder, prior to assembly to loaded delay holder (see dwg. 8853675).

Categories	Defects	Method of inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Varnish or foreign matter on concave surface or in flash hole of primer anvil	Visual	01001
102.	Evidence of poor workmanship (see 3.4)	Visual	01002
Minor:	None defined.		

4.3.2.2 Delay holder, prior to assembly to loaded primer holder (see dwg. 8853675).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL .25 percent		
101.	Evidence of foreign matter in relay cavity	Visual	02001
102.	Evidence of foreign matter on relay charge	Visual	02002
Minor:	AQL .40 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	02003

4.3.2.3 Assembly (see dwg. 8853675).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Total length, max.	Gage	03001
102.	Outside diameter of holder, max.	Gage	03002
103.	Primer missing or inverted	Visual	03003
104.	Relay missing or inverted	Visual	03004
105.	Crack in relay crimp	Visual	03005
106.	Thread damaged	Visual	03006
107.	Explosive exposed	Visual	03007
Minor:	AQL 2.5 percent		
201.	Concentricity of delay holder with thread	Gage	03008
202.	Depth to primer, minimum	Gage	03009
203.	Evidence of poor workmanship (see 3.4)	Visual	03010
204.	Coating on relay missing or incorrect	Visual	03011
205.	Varnish on primer cup	Visual	03012

4.3.2.4 Carton, sealed (inner and outer) (see dwg. 7548278).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Assembly damaged so that contents are exposed or liable to become exposed	Visual	04001
102.	Sealing strip incomplete, badly wrinkled, or not in contact with surface for at least ½ inch around entire carton	Visual	04002
Minor:	AQL 1.0 percent		
201.	Contents move when carton is shaken	Manual	04003
202.	Marking misleading or unidentifiable	Visual	04004

4.3.2.5 Bag, barrier, sealed (see dwg. 7548278).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		

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Categories	Defects	Method of inspection	Code No.
Major:	AQL 0.25 percent		
101.	Waterproovness destroyed by rip, tear, puncture, or separation of heat seal	Visual	05001
Minor:	None defined.		

4.3.2.6 Sealed wood packing box (see dwg. 7548277).

Categories	Defects	Method of inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Strapping missing, broken or loose	Visual-Manual	06001
102.	Contents exposed	Visual	06002
103.	Board broken or split	Visual	06003
Minor:	AQL 1.50 percent		
201.	Strapping improperly assembled	Visual	06004
202.	Marking misleading or unidentifiable	Visual	06005
203.	Contents loose	Manual	06006

4.3.3 *Testing.*

4.3.3.1 *Heat seal test of barrier bag.* (See dwg. 7548278)—Major defect—Code No. 07001. The sampling plan for this test shall be in accordance with Standard MIL–STD–105D, table II–A, Code letter H, and AQL 0.25 percent. The test shall be performed as specified in 4.4.1.

4.3.3.2 *Check test for possible deterioration of Primer, Percussion, M54.* (See Specification MIL–P–13392)—Code No. 02004. If the total elapsed time between original acceptance of the M54 primer lot and the assembly of that lot into the delay element exceeds two years, or if the M54 primers have been subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in Specification MIL–P–13392 immediately before the M54 primer lot is assembled into the elements. The test shall be performed by the contractor on assemblies selected by the Government inspector at the facility assembling the delay element (see 6.4).

4.3.3.3 *Functioning.* (See 3.3)—Major defect—Code No. 03013.

4.3.3.3.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified herein, the sampling plan for this test shall be in accordance with Standard MIL–STD–105D, table III–A, Code letter M and AQL of 0.25 percent. The test shall be performed as specified in 4.4.2 using test equipment in accordance with 4.3.4.

4.3.3.3.2 After three consecutive lots have met the criteria of 4.3.3.3.1 the sampling plan for this test shall be in accordance with Standard MIL–STD–105D, table III–A, Code letter L and AQL of 0.25 percent. The test shall be performed as specified in 4.4.2, using test equipment in accordance with 4.3.4.

4.3.4 *Inspection equipment.* Index of Inspection Equipment Number IEL–8853675 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of

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items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL–I–45607.

4.3.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures.

4.4.1 *Heat seal test of barrier bag.* The test procedure for this test shall be as prescribed in Specification MIL–P–116 for the "Heat Seal Test."

4.4.2 *Functioning.* The delay element assembly shall be tested by detonating the M54 percussion primer by means of a steel firing pin having a point of 28 degrees minus 2 degrees which has a .015–.004 flat with a .004 maximum radius. A steel ball weighing 7.94 plus or minus .02 ounces shall be dropped from a height of 10 inches on the firing pin. The height shall be measured from the bottom of the steel ball to the top of the firing pin. Observation shall be made for compliance with the applicable requirements.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level C.* The delay element assemblies shall be preserved and packaged in accordance with dwg. 7548278.

5.2 Packing.

5.2.1 *Level C.* The delay element assemblies, preserved and packaged as specified in 5.1.1, shall be packed in accordance with dwg. 7548277.

5.3 Marking. Marking shall be in accordance with dwgs. 7548277 and 7548278.

5.4 Data cards. Data cards shall be prepared for each lot in accordance with the information specified in Specification MIL–A–2550.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this Specification.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (00 Form 1242–3) the numbers of drawings of existing equipment designs or codes as indicated in 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "contractor design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5, and 6.3.6 will be designated in the Inspection Equipment List by the following codes:

CDOF —Contractor's design responsi-

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bility on Ordnance format; ORDM–608–12.

CDCF —Contractor's design responsibility on contractor format.

OD —Ordnance design.

ODMU —Ordnance design, mandatory for use.

CE —Commercial equipment.

SCD —Specification control drawing.

6.3.3 *Ordnance designs.* Ordnance designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and nonmandatory. When an Inspection Equipment List references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with nonmandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawings.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDOF shall comply with the format and requirements of ORDM 608–12, and, in addition, contain the detail and information specified above.

6.3.7 *Submission to contractor designs.* Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA–ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Cost of check test. The contracting officer will arrange for the contractor to be re-

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imbursed for the expense incurred in performance of the check test. The test shall be conducted at Government expense without cost to the contractor who loaded the percussion elements or to the contractor assembling the percussion elements into the primer assemblies and shall not constitute a basis of rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the percussion element lots were under the jurisdiction of either contractor.

6.5 Check test for possible deterioration of delay elements. The following test is designed to prevent the loading into a major ammunition-item of delay elements which have deteriorated. The test shall be performed on each lot of delay elements which have been stored under normal conditions for more than two years, or under adverse conditions for any period of time. If the specification for the ammunition item into which the delay element is loaded does not require that the check test for deterioration be performed, the contracting officer shall insert a provision for the performance of the check test in the contract (see 6.4).

6.5.1 *Functioning.* (See 3.3.) The sampling plan for this test shall be in accordance with Standard MIL–STD–105D using code letter L and an AQL of 0.40 percent. Any delay element assembly which fails to comply with the specified requirements shall be classed defective. The test shall be performed as specified in 4.4.2 using equipment in accordance with 4.3.4. Rejection shall be in accordance with Standard MIL–STD–105D.

6.6 Supersession. This specification supersedes, in part, Military Specification MIL–D–13512 (Ord), dated 24 June 1954 which included both M1 and M2 delay elements. (The loading, assembling and packing of the M2 delay element is now covered by Military Specification MIL–D–46486.)

6.7 Delay time. Delay time is the length of time from the instant the firing pin strikes the primer to the functioning of the relay.

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